GATE

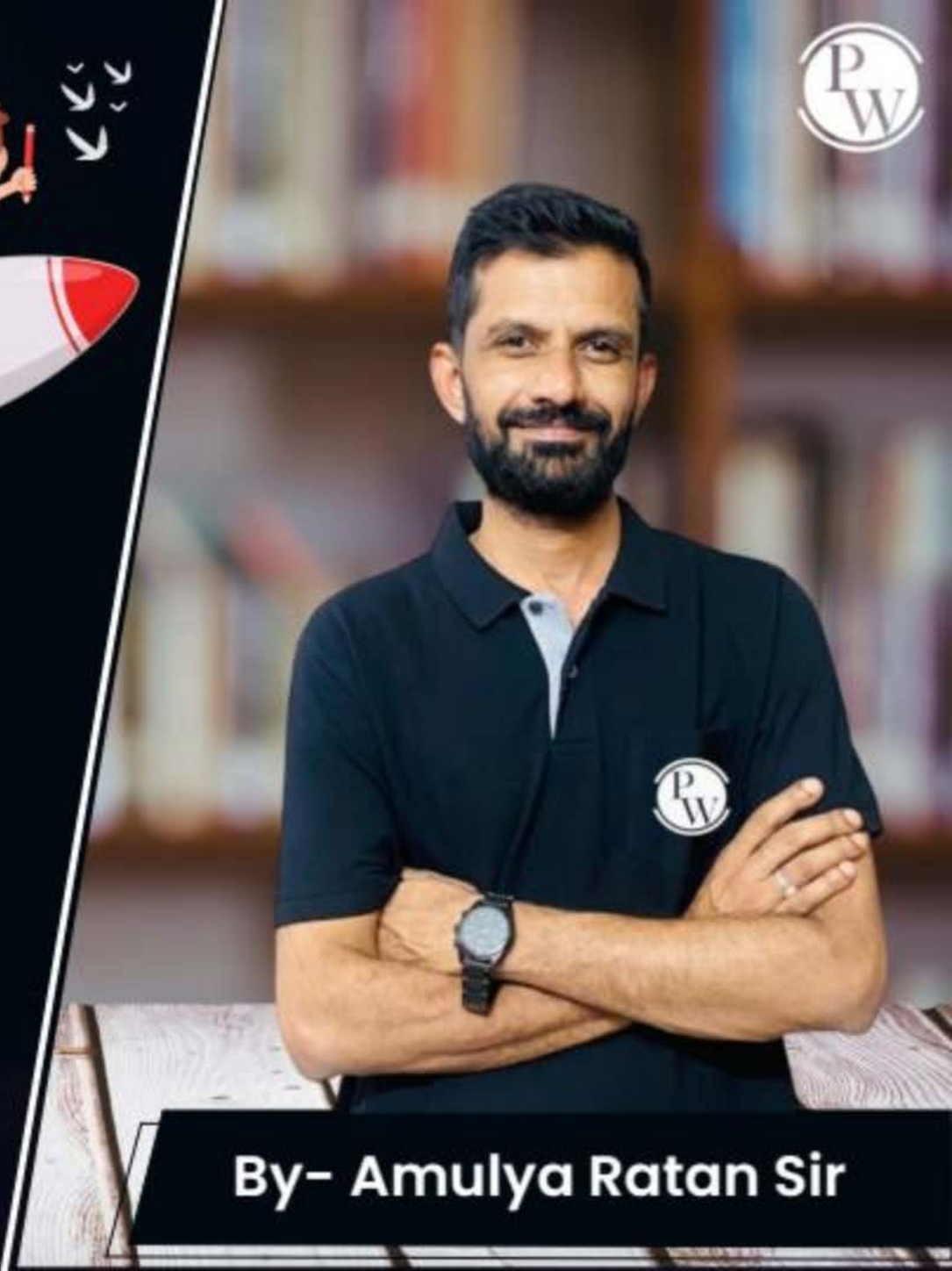
ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 11



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Time & Work

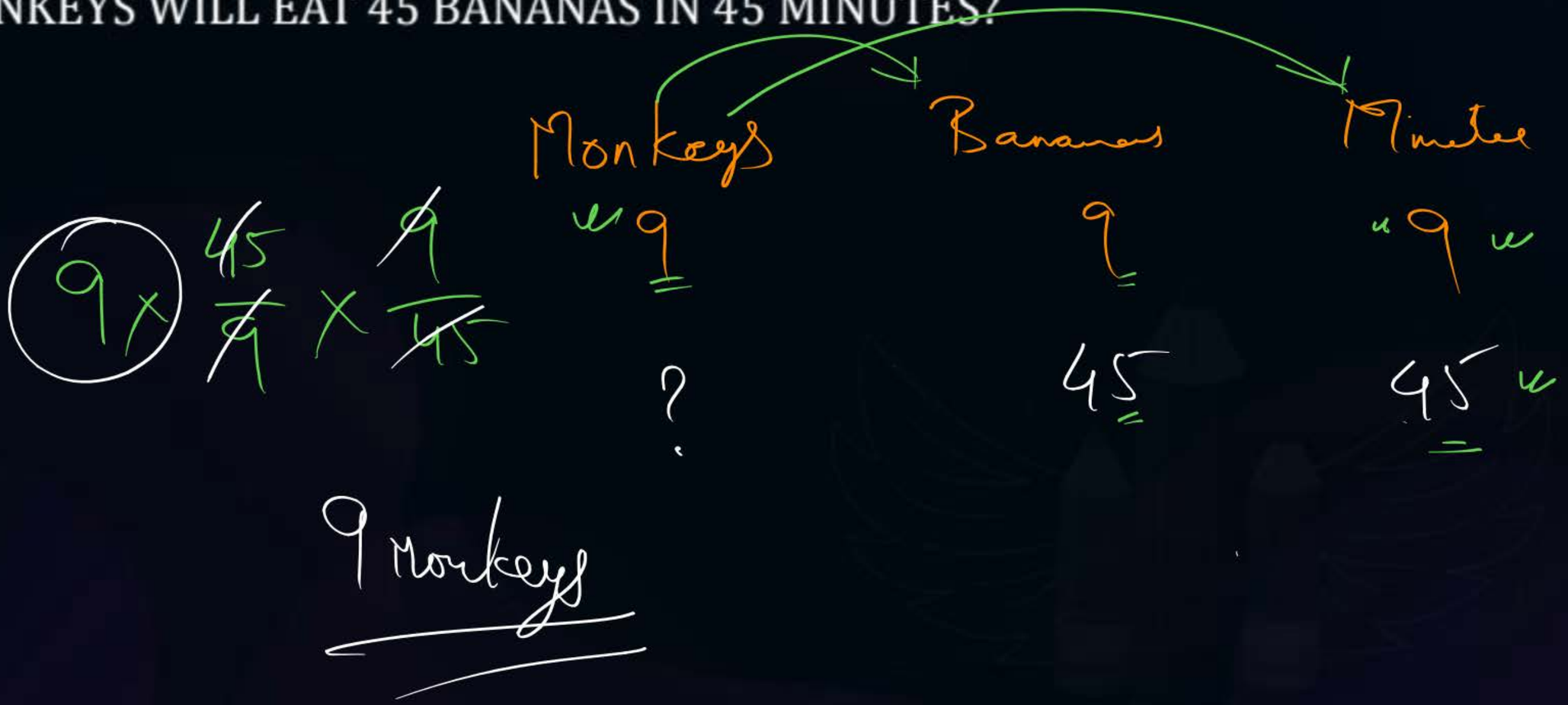
Topics to be Covered



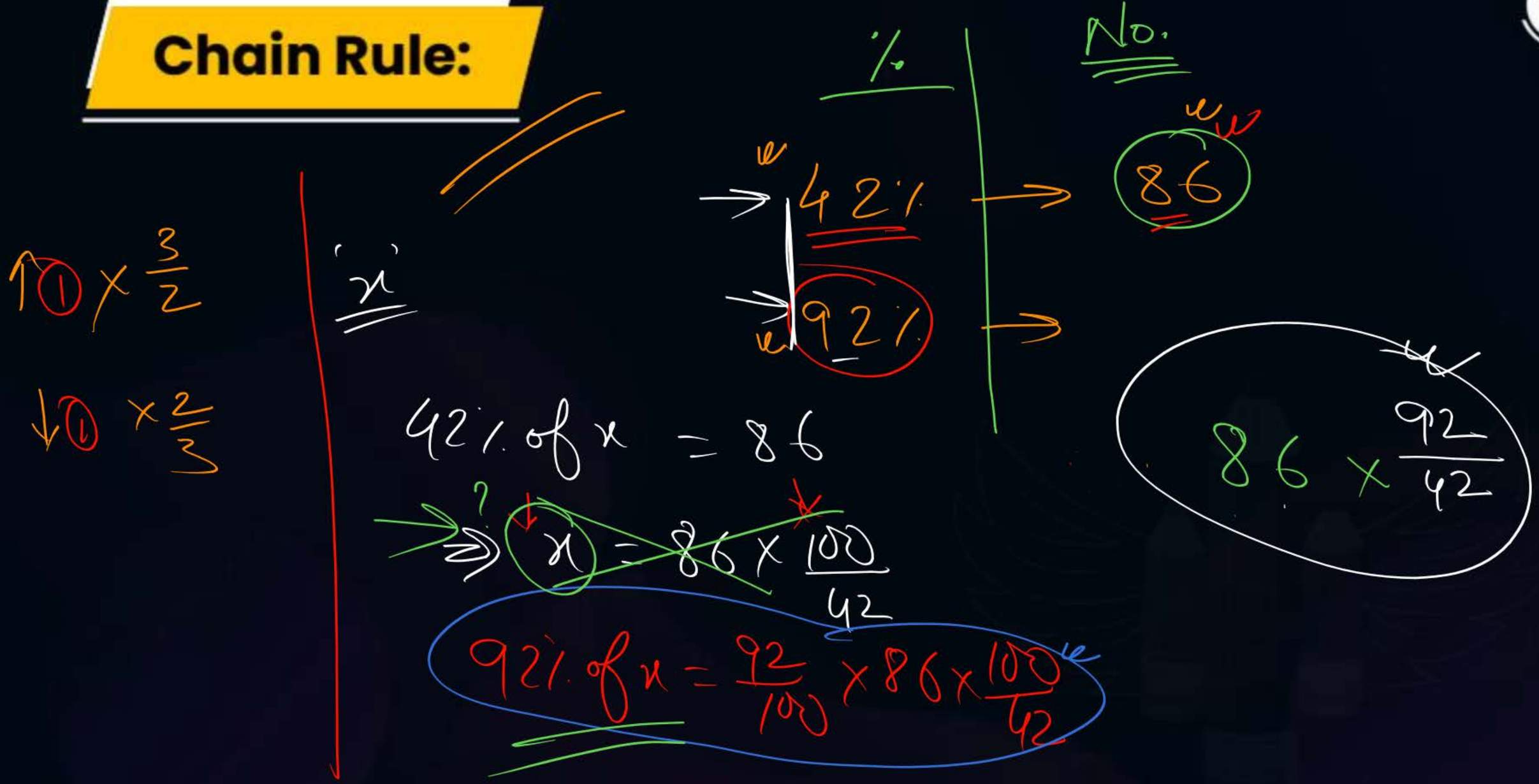
Topic-1 Pipes and Cistern

[MCQ]

#Q. If 9 MONKEYS EAT 9 BANANAS IN 9 MINUTES, THEN HOW MANY MONKEYS WILL EAT 45 BANANAS IN 45 MINUTES?



Chain Rule:


%
No.
42%
92%
86
x
↑ ① × 3/2
↓ ① × 2/3
42% of x = 86
⇒ x = 86 × 100/42
92% of x = 92/100 × 86 × 100/42
86 × 92/42

[MCQ]

#Q. 12 men can do a work in 15 days working 8 hours a day. In how many days can 9 men do the same work, working 10 hours a day?



[MCQ]

Extra days ⇒ 120

#Q. A hostel with 600 students had sufficient food for 210 days. After 30 days, 240 students left the hostel. Now the remaining food will last for how many days?

Total Food

300 + 30

= 330 days

After 30 days

Students	Days
600	180 (210 - 30)
(600 - 240) 360	?

$1 \times \frac{\cancel{600}^{300}}{\cancel{360}_{2}} = 1 \times \frac{300}{2} = 150$

Remaining Food ⇒ 300 days

[MCQ]

6 am + 6 hrs = 12 PM

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If that clock represents 1 pm the same day, what must be the correct time?

Chain Rule

$6\phi \times \frac{\pi^c}{7\phi}$

$= 6 \text{ hrs}$

C.W.	W.W
60	70
?	7 hrs $\times$ 60 e

60

[MCQ]

#Q. A Clock which gains 10 minutes in every one hour was set correct at 6 am. If the correct time is 12 pm the same day, what would be the time shown by that clock??



[MCQ]



$\frac{2}{3}$

#Q. A contractor under takes to make a road in 40 days and employs 25 men. After 24 days, he finds that only one-third of the road is made. How many extra men should he employ so that he is able to complete the work 4 days earlier?

A 75

B 100

C 50

D None of these

100 − 25 = 75

Men	Days	Work
25	24	$\frac{1}{3}$
?	36 − 4 = 32	$\frac{2}{3}$

100 Men

$25 \times \frac{24}{32} \times \frac{2}{3} \times \frac{3}{1} = 100$

[MCQ]

#Q. A can do a work in 60 days whereas B in 20 days. If they together took a contract of ₹32000, then what would be the share of A?

(Days)

A : B = 60 : 20 = 6 : 2 = 3 : 1

(Eff.) A : B = 1 : 3

$\frac{1}{4} \times 32,000 = 8000$

A ₹ 8000

B ₹ 6000

C ₹ 1800

D ₹ 2040

Salary

Wages

₹ 6000

A : B = 5 : 7 (Days)

A : B = 7 : 5 (Eff.)

A : B = $\frac{1}{10}$: $\frac{1}{20}$

2 : 1

A : B = 1 : 5 (Eff.)

A : B = 5 : 1 (Days)

A = 10
B = 20
} Days
A : B = 10 : 20
= 1 : 2

A : B : C = 1 : 2 : 3 (Days)

A : B : C = 6 : 3 : 2 (Eff.)

$\frac{1}{x}$ $\frac{1}{y}$

$y : x$

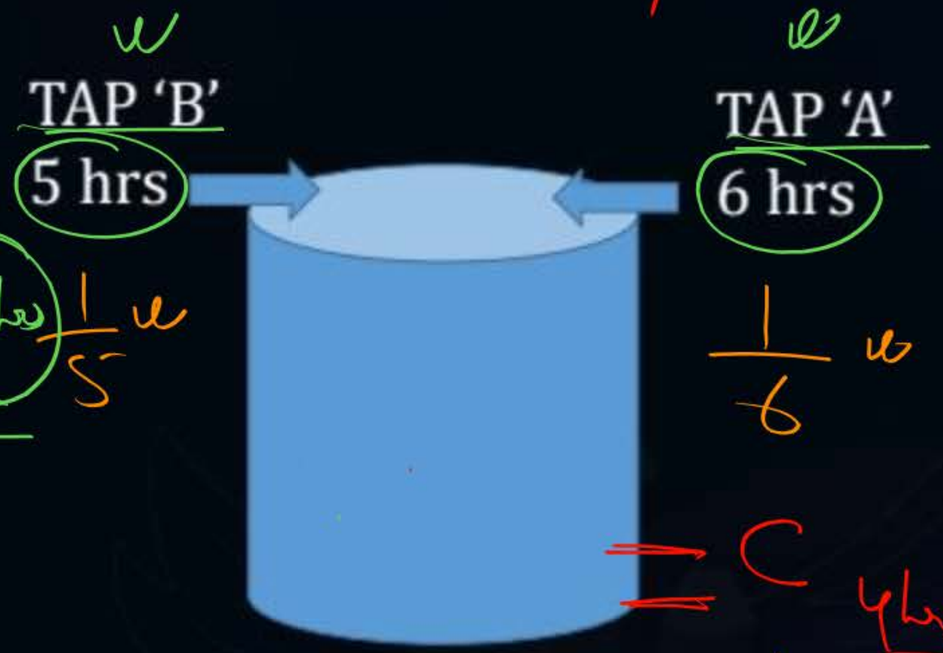


Pipes & Cistern

Tank

Pipes & Cistern

Empty (−)
filling (+)



$\frac{1}{6} + \frac{1}{5} = \frac{11}{30}$ → $\frac{30}{11} = 2\frac{8}{11}$ hrs

$\frac{1}{5}$ u

$\frac{1}{6}$ u

C = 4 hrs

$-\frac{1}{4}$

A & B & C ⇒ $\frac{1}{6} + \frac{1}{5} - \frac{1}{4}$

$= \frac{10 + 12 - 15}{60} = \frac{7}{60}$ → $\frac{60}{x} = 8\frac{4}{7}$ hrs

[MCQ]

$A = \frac{1}{8}$ | $B = \frac{1}{12}$ u

#Q. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively. Both the pipes are opened together and after 3 minutes, pipe A is turned off. What is the total time required to fill the tank?

'x' minutes



A 3 minutes

B 7.5 minutes

C 10 minutes

D 4.5 minutes

$$\frac{x}{12} + \frac{3}{8} = 1$$

$$\Rightarrow 2x + 9 = 24$$

$$\Rightarrow 2x = 15$$

$$\therefore x = \frac{15}{2} = 7.5 \text{ min}$$

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

A 4.5 hours

B 2 hours

C 6 hours

D 3.2 hours

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

A 30

B 25

C 20

D 15

[NAT]

$A = \frac{1}{20}$ | $B = \frac{1}{12}$ | $C = \frac{-1}{18}$

#Q. A tank consist of three taps A, B and C. Tap A and tap B can fill the tank in 20hrs and 12 hrs respectively, whereas tap C can empty the tank in 18hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A.....so on. In how many hours the tank would be filled?

1st hr ⇒ A&C

$= \frac{1}{20} - \frac{1}{18} = \frac{9-10}{180} = \frac{-1}{180}$

2nd hr ⇒ B&C

$\frac{1}{12} - \frac{1}{18} = \frac{5}{180}$

$\frac{1}{36}$ 2nd & 3rd 2 hrs

1	2	3	4	5
A & C	B & C	A & C	B & C	A & C

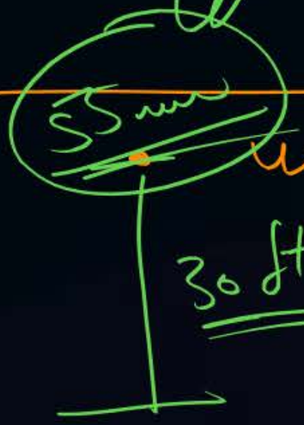
6 hours

2nd & 3rd

$89\frac{4}{5}$ work

$\frac{5}{180} - \frac{1}{180} = \frac{4}{180} = \frac{1}{45}$ ⇒ 2 hours

44 ×4 = 44 ×4



$\frac{1}{45}$?

2 minutes ⇒ 1 ft.

54 minutes ⇒ 27 ft.

56 minutes ⇒ 28 ft

44 ⇒ 88 hours

$\frac{44}{45}$

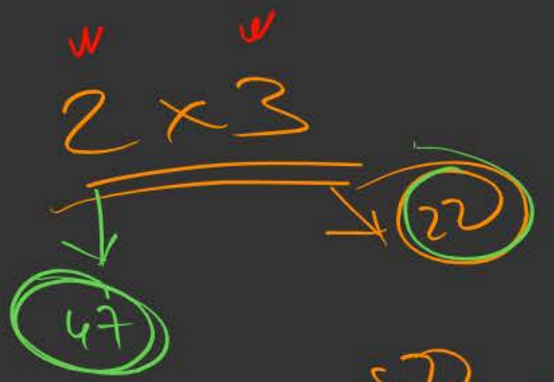
+1

89 hours

1 work B & C = 36 hrs

$\frac{1}{45}$ = B & C = $\frac{1}{45} \times 36 = \frac{4}{5}$ hrs

$50! \rightarrow 6$ (22)



$\frac{50}{2} \rightarrow 25$

$\frac{25}{2} \rightarrow 12$

$\frac{12}{2} \rightarrow 6$

$\frac{6}{2} \rightarrow 3$

$\frac{3}{2} \rightarrow 1$

$\frac{1}{2} \rightarrow 0$

$\frac{32}{2} \rightarrow 16$

$\frac{16}{2} \rightarrow 8$

$\frac{8}{2} \rightarrow 4$

$\frac{4}{2} \rightarrow 2$

$\frac{2}{2} \rightarrow 1$

$\frac{1}{2} \rightarrow 0$



2 mins Summary

Topic Chain Rule

Pipes and Cistern

THANK - YOU